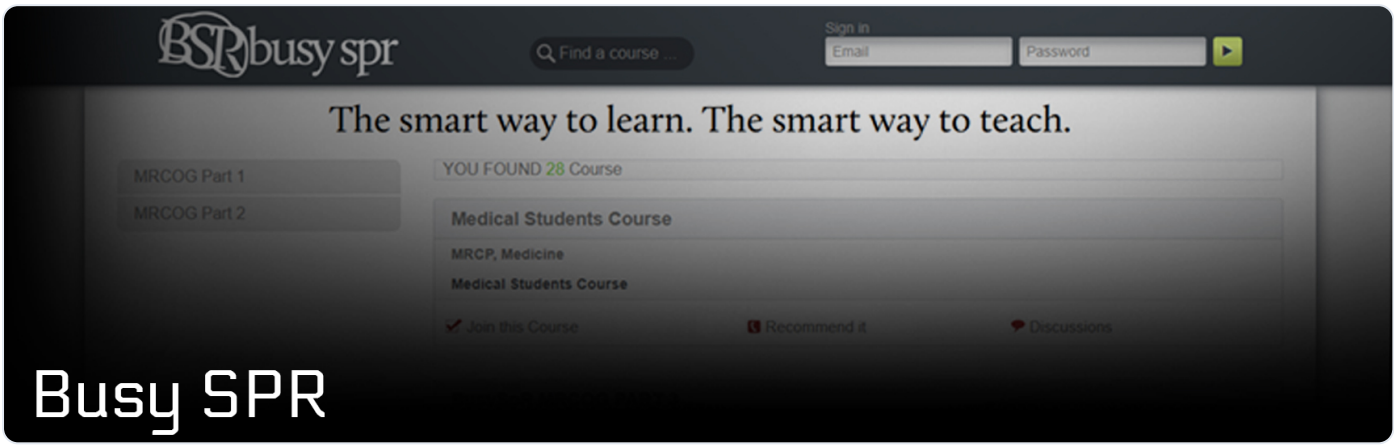
EDUCATIONAL OR EDTECH PROJECTS VIDEOS

Busy SPR

BusySPR — Postgraduate Medical Examination & Registrar Revision Platform



1. Executive Summary & Overview

BusySPR is an online medical education and e-learning portal engineered for junior doctors, medical registrars (SpRs), and healthcare professionals preparing for high-stakes postgraduate medical qualifications — primarily MRCOG Part 1 & Part 2, DRCOG, and specialty medical board exams. The platform integrates authentic UK Royal College question formats, timed mock testing engines, structured clinical study notes, peer/mentor discussion boards, and paid subscription access powered by PayPal.

2. PROBLEM & ARCHITECTURAL CHALLENGE

The Problem: Postgraduate medical examinations like MRCOG and MRCP are among the toughest professional licensing tests in the world, with pass rates often under 40%. Candidates struggle with scattered and outdated revision materials, lack of authentic Royal College question formats, and lack of trusted clinical discussion channels for exam question recalls.

The Challenge: Replicating authentic Royal College examination logic (Best of Five single best answer, Extended Matching Questions with shared scenario stems, N of M multiple true/false, timed mocks with negative marking options), multi-tenancy for international medical colleges across the UK and West Africa, secure paid subscription gating with PayPal, and expert contributor royalty/mass-payout tracking.

3. THE ENGINEERING SOLUTION

An end-to-end medical learning management system featuring customizable mock testing algorithms, question banks categorized by clinical specialty, interactive discussions for disputed question recalls, high-yield clinical notes and multimedia figure libraries, automated PayPal subscription processing, and multi-tenant subdomains for partner universities.

4. Key Functional Modules

Homepage & Course Discovery

Student Registration & Security (/user/add.html)

Top rated courses, recent clinical discussions, smart search, and registration.

Secure account creation with international country localization and dynamic security captcha.

Course Directory & Subscription Pricing (/?comp=subscription&cid=14)
Course curriculum details, free trial enrollment, and multi-month subscription packages with PayPal checkout integration.

Best of Five (BOF5) Exam Engine
Royal College standard clinical scenarios where candidates select the single most appropriate clinical intervention.

Extended Matching Questions (EMQ) Testing System
Linked theme-based medical cases with shared diagnostic/treatment option lists.

Multiple Choice Questions (MCQ) Timed Mocks
Randomized question pooling from the question bank with detailed answer rationales.

Clinical Revision Notes & Radiology Figures
High-yield clinical study guides, medical figures, and diagnostic audio clips.

Peer & Mentor Discussion Forum
Threaded case discussions with moderation and specialist pinned posts.

Admin & Faculty Panel (/panel/)
Question authoring, candidate enrollment reporting, and PayPal MassPay royalties distribution.

5. Technology Stack & Deployment Architecture

Backend: PHP (Custom MVC Micro-Framework)

Database: MySQL / MariaDB (Optimized question relational schema)

Frontend: Server-Side Rendered PHP, jQuery, jQuery UI, CSS3

Payments & Integrations: PayPal Checkout SDK, PayPal MassPay, PHPMailer / SMTP

Architecture: Multi-Tenant Subdomain Configuration with Centralized Question Engine